

TradeBot Current System Design Specification

TradeBot 現行システム設計仕様書

```
**Version:** 2026.07.12
**Generated (JST):** 2026-07-12T06:32:15+09:00
**State:** PAPER TRADE ONLY — REAL ORDERS NOT AUTHORIZED / NOT IMPLEMENTED
**Production YAML:** `configs/small_paper_pilot_q070_cap3_entry_price_risk_guard_trailing_mfe_shadow.yaml`
**YAML SHA256:** `a20e40ed1bf52624478ecfecf73270a2e3f8df293b37ebcf9d5534ba410e4690`
**Source of Truth:** Runtime BAT/PS1/Python/YAML (not historical Phase prose)
```

1. 文書概要

本書は 2026-07-12 時点の TradeBot Paper Runtime を、実コード到達性に基づき正式仕様化した設計書である。過去 Phase 文書は参考に留め、現行 `run\_paper\_trade\_checked.bat` 起動経路と production YAML を優先する。

2. システム目的

- ・日本株デイトレ候補を Paper（仮想）で評価し、ENTRY/EXIT・Summary・Discord・Capture Seal・W4S を運用する。
- ・市場 PUSH を Capture Sidecar で独立保存し、Paper 障害でも当日テープを保全する。
- ・実注文は未実装・未許可。Safety フラグと HARD_FAIL で実発注経路を封じる。

3. 対象範囲

- ・Checked Runner 起動～Capture ONLINE～Paper AM/PM～W4S～Capture 15:35 finalize
- ・production YAML の ENTRY/EXIT/Universe/Refresh/Discord/Seal
- ・Runtime Gate 28 ノード契約

4. 対象外

- ・実注文送信・口座振替・本番資金移動
- ・research_long フル期間リプレイ（Monday Gate 除外）
- ・廃止 Shadow（Exit Shadow Monitor, rise5 shadow, VWAP shadow reject）の再有効化前提

5. 全体アーキテクチャ

A. 起動構成図

[Mermaid diagram 1]

```
flowchart TD
    BAT[run_paper_trade_checked.bat] --> PS1[run_paper_trade_checked.ps1]
    PS1 --> CR[paper_trade_checked_runner]
    CR --> UNI[universe resolve]
    CR --> REG[registration coordinate]
    CR --> CAP[spawn Capture Sidecar]
    CR --> PRE[prechecks cache/preflight/smoke/recovery/safety]
    CR --> PBAT[run_paper_trade.bat]
    PBAT --> PRE2[live pipeline preflight + smoke]
    PBAT --> AMR[run_core10_dynamic40_am_pm_daily_runner]
    AMR --> AM[AM session pilot_runner]
    AMR --> PM[PM session pilot_runner]
    CR --> W4S[phase687w4s forward soak]
    CR --> CFIN[capture finalize verify / 15:35]
    CAP --> SUP[market_capture_supervisor max restart 1]
    SUP --> SID[market_capture_sidecar]
```

B. Process 図

- **Checked Runner** (`paper\_trade\_checked\_runner`) — 親オーケストレータ
- **Paper Runtime** (`pilot\_runner` via AM/PM daily runner) — ENTRY/EXIT
- **Market Capture Sidecar** — 別 PID、15:35 まで
- **Discord Worker** — async fail-open
- **Kabu WebSocket** — Paper + Capture (dual WS 公式保証は未確定)
- **File Writer** — capture part JSONL / paper events / seals

C. データフロー図

[Mermaid diagram 2]

```
graph LR
    subgraph PaperPath
        KP[Kabu PUSH] --> PE[Paper Evaluation]
        PE --> EN[ENTRY]
        PE --> EX[EXIT]
        EN --> POS[Position]
        EX --> POS
        POS --> EV[Event JSONL]
        EV --> SUM[Canonical Summary]
        SUM --> DC[Discord W10 Router]
    end
    subgraph CapturePath
        KP2[Kabu PUSH] --> IC[Independent Capture]
        IC --> JL[push_part_*.jsonl]
        JL --> MAN[capture_manifest / status]
        MAN --> SEAL[capture_seal 15:35]
    end
```

D. Session lifecycle 図

[Mermaid diagram 3]

```
stateDiagram-v2
    [*] --> START
    START --> PREFLIGHT
    PREFLIGHT --> CAPTURE_ONLINE
    CAPTURE_ONLINE --> AM
    AM --> AM_FINALIZE
    AM_FINALIZE --> PM
    PM --> PM_FINALIZE
    PM_FINALIZE --> W4S
    W4S --> CAPTURE_1535_FINALIZE
    CAPTURE_1535_FINALIZE --> [*]
```

E. Registration ownership 図

- Paper は registration **owner/follower 協調**下で差分 refresh。
- Capture active 中: `safe\_paper\_unregister` が `unregister\_all` を defer (W11A)。
- Capture reconnect: `clear\_first=false` (registration 維持)。
- Code: `registration\_lifetime.safe\_paper\_unregister` L306-359
- Sidecar は manifest follower — `unregister\_all` 禁止。

6. 起動シーケンス

1. `run\_paper\_trade\_checked.bat` → PowerShell Bypass → `run\_paper\_trade\_checked.ps1`
2. `python -m small\_paper.paper\_trade\_checked\_runner` (`PaperTradeCheckedRunner.run` L1669-1801)
3. dotenv / disk / kabu readonly / universe / registration / **Capture start + ONLINE wait**
4. Paper prechecks (cache, preflight, smoke, recovery, design, safety flags)
5. `run\_paper\_trade.bat` を一度だけ起動
6. BAT 内: live pipeline preflight → production smoke → `run\_core10\_dynamic40\_am\_pm\_daily\_runner.py --universe-mode core10-dynamic40-price-risk-filter-shadow --enable-intraday-refresh --exit-policy-shadow trailing-mfe`
7. AM → PM → 戻って W4S → Capture finalize verify (live は 15:35 継続可)

7. Process / Thread構成

Process	PID関係	備考
Checked Runner	親	precheck+orchestrate
Capture Sidecar	別PID (supervisor配下)	Paper失敗独立、restart<=1
Paper BAT → AM/PM runner → pilot	子	AM/PM順次
Discord async worker	Paper/Checked内スレッド	fail-open
Capture writer thread	Sidecar内	queue/overflow

8. ディレクトリ構成

詳細は `tradebot\_directory\_map.md`。主要:

- `kabu\_native/src/small\_paper/` runtime
- `kabu\_native/configs/` production YAML
- `kabu\_native/data/market\_capture/YYYYMMDD/` Capture
- `kabu\_native/results/small\_paper/` Paper artifacts
- `kabu\_native/runtime/` registration lock/manifest
- `docs/current\_system\_design/` 本書

9. 環境変数と設定ファイル

- Config SoT: `configs/small\_paper\_pilot\_q070\_cap3\_entry\_price\_risk\_guard\_trailing\_mfe\_shadow.yaml` (sha256=`a20e40ed1bf52624478ecfecf73270a2e3f8df293b37ebcf9d5534ba410e4690`)
- Pin: `configs/production\_config\_sha256.pin`
- `.env` via `small\_paper.env\_loader` (Webhook URL はログに出さない)
- 詳細: `tradebot\_environment\_variables.md` / `tradebot\_config\_reference.csv`

10. Universe設計

- Mode: `core10-dynamic40-price-risk-filter-shadow` (BAT 固定)
- Core10 + Dynamic40、**最大50** (Kabu PUSH 登録上限)
- Price-risk filtering 適用
- Open position carry 優先、previous subscription keep on degraded refresh
- `open\_symbols\_exceed\_cap` → CONTINUE (`will\_stop=false`)

11. Kabu Station接続設計

- 前提: Kabu Station 起動・ログイン・API 利用可
- Readonly readiness: checked runner `step\_kabu\_readonly`
- Paper PushSource default: Kabu direct WS
- Capture: preferred PASSIVE_DUAL_WEBSOCKET (公式保証は未確定)

12. WebSocket / Registration設計

- Registration SoT: `runtime/market\_registration\_manifest.json` + lock
- Refresh: generation 付き差分更新
- Capture active: Paper `unregister\_all=0`、reconnect `clear\_first=false`
- Code: `market\_capture\_registration.py`, `registration\_lifetime.py`

13. Market Capture Sidecar設計

- 別 PID / Paper failure 独立 / **15:35 JST** finalize
- Supervisor `MAX\_AUTO\_RESTARTS=1` (`market\_capture\_supervisor.py`)
- Part rotation: `max(existing)+1`、`O\_CREAT|O\_EXCL` (`market\_capture\_writer.py`)
- heartbeat / PID file / queue overflow → dropped_event / DEGRADED
- registration mismatch / sequence / manifest / seal / metrics
- disk full / malformed PUSH / reconnect / finalize
- **W11A**: Paper は Capture active 中 unregister しない ; Windows PID probe は OpenProcess

14. Paper Runtime設計

- Core: `pilot\_runner` push pipeline → ExposureGate → Observer positions
- Gate: `ExposureGate.evaluate\_entry` L235-698
- paper_only=true, dry_run=true, shadow_only=true (YAML)
- live_trading_enabled=false, order_enabled=false

15. ENTRY設計

条件名	status	config key	現行値	入力	判定
PBv2 profile / entry_score_v2	MAINLINE_ACTIVE	entry_profile / entry_score_v2_min / momentum_score_cutoff_max	momentum_volume_v2 / min=3 / cutoff<=0.2546	entry_expectancy_score_v2, momentum_continuation_score, board tier	score_v2>=min AND momentum_score_cutoff_pass AND board_mid_or_high
Momentum low (explicit cutoff)	MAINLINE_ACTIVE	momentum_score_cutoff_max	0.2546	momentum_continuation_score	score <= cutoff (Phase472 PBv2)
Board mid/high required	MAINLINE_ACTIVE	(derived with entry_score_v2)	board_mid_or_high_required_for_v2	board imbalance / board tier tokens	board mid or high required for v2 accept
OR Open Strength Overlay	MAINLINE_ACTIVE	or_overlay_enabled / cap_pbv2 / cap_or / or_max_update_count	true / 4 / 1 / 8	OR open strength features, update_count	CAP_SPLIT_4_1: PBv2<=4, OR<=1, total<=5
Price Risk Guard	MAINLINE_ACTIVE	entry_price_risk_guard_enabled / min_entry_price / max_tick_ratio_pct / apply_mode	true / 50.0 / 5.0% / reject_entry	entry price, tick size ratio	price>=min AND tick_ratio<=max; apply reject_entry
High Drift Pullback	MAINLINE_ACTIVE	high_drift_guard_enabled	true	day_high distance, r5/r10/r15, dynamic40	dynamic40 AND ((dh>=1.2% AND r10<-0.15% AND r5>r10) OR (dh>=1.5% AND (r15<-0.5% OR r5<-0.5%)))
Weak Shape	MAINLINE_ACTIVE	weak_shape_reject_enabled	true	opening_peak / slow_opening_peak shape labels	reject opening_peak / slow_opening_peak at ENTRY
Late Chase Guard	MAINLINE_ACTIVE	late_chase_guard_enabled	true	r10, day_high_distance	r10<0.3719 AND day_high_distance<1.1872 → reject
Classic RSI late chase	MAINLINE_ACTIVE	classic_late_chase_rsi_guard_enabled / threshold	true / 80.0	late_chase_cluster flag, RSI14	late_chase_cluster AND RSI14>=threshold
Reentry RSI	MAINLINE_ACTIVE	reentry_rsi_guard_enabled / threshold	true / 60.0	prior stop_hit, RSI14	re-entry after stop_hit requires RSI14>threshold
Entry Quality G9	MAINLINE_ACTIVE	entry_quality_guard_enabled / max_spread_bps / max_update_count	true / 50.0 / 5	spread_bps, update_count	require spread<=50bps AND update_count<=5
Entry Cluster Guard	MAINLINE_ACTIVE	entry_cluster_guard_enabled / reject_clusters / exception / liquidity_burst	true / clusters=[5] / csubs=[] / exception=true / thr=0.052267	cluster model features, liquidity_burst	reject cluster5 unless E4 liquidity_burst exception
Flat-band mainline	MAINLINE_ACTIVE	pbv2_flat_band_mainline_enabled (+ threshold keys)	true; rise5[0.0,0.5] rise10[-0.5,0.5] overheat>=2.0	entry_rise_5min_pct, entry_rise_10min_pct, PBv2 pool	flat band + overheat evaluate_flat_plus_overheat
Near day-high + low momentum (Dynamic40)	MAINLINE_ACTIVE	enable_near_day_high_low_momentum_dynamic40_guard	true	dynamic40, day-high proximity, momentum	production ENTRY reject for D40 near-high low-mom
Position cap	MAINLINE_ACTIVE	max_concurrent_positions / position_cap_mode / position_cap_release / cap_pbv2 / cap_or	5 / true / structural_exit / 4/1	open position count by pool	total<=5 with OR split; release on structural exit

条件名	status	config key	現行値	入力	判定
same_symbol_open_policy	MAINLINE_ACTIVE	same_symbol_open_policy	no_overlap_replace	open positions for symbol	reject ENTRY while same symbol open (no overlap replace chain)
Allowed trading windows	MAINLINE_ACTIVE	allowed_trading_windows / use_market_time_window	09:05-11:23, 12:33-15:20; use_market_time_window=true	market clock JST	outside window → reject
Stale price / board freshness	MAINLINE_ACTIVE	entry_freshness_guard_enabled / entry_max_price_age_sec / entry_max_board_age_sec / freshness_semantics_v2	true / 3.0s / 3.0s / v2=True	CurrentPriceTime, board update time, event age	price/board age <=3s; trade_stale tag_only at 10s
Daytrade suitability	MAINLINE_ACTIVE	daytrade_suitability_enabled / rule / apply_mode	true / volatility_liquidity_top50 / reject_entry	vol/liq ranking prior sessions	reject if not in suitability set
Stop Low MFE Guard	NOT_RUNTIME_REACHABLE	stop_low_mfe_guard_enabled	false	n/a (disabled)	gate branch not taken when false
Daily loss / risk cluster	MAINLINE_ACTIVE	daily_loss_guard_enabled / daily_loss_guard_pct / risk_cluster_*	true / -2.5% / consecutive=5	session PnL, consecutive losses	block further ENTRY when tripped

candidate → accept → reject

- PUSH → candidate trade 構築
- freshness / scan batch / same-symbol / universe membership
- `ExposureGate.evaluate\_entry`（上記表の順で reject 可）
- OR overlay / position cap 分割
- accept → observer open + Discord TRADE_ACTUAL ; reject → JSONL + 条件により CAP_BLOCKED

reject reason 一覧（本線）

`momentum\_low\_required`, `entry\_score\_v2\_below\_threshold`, `entry\_price\_risk\_guard`, `high\_drift\_pullback`, `weak\_shape\_reject`, `late\_chase\_guard`, `classic\_late\_chase\_rsi\_over80`, `reentry\_rsi\_guard\_below60`, `entry\_quality\_guard\_spread`, `entry\_quality\_guard\_update\_count`, `entry\_cluster\_guard`, `flat\_band\_mainline`, `near\_day\_high\_low\_momentum\_dynamic40\_guard`, `daytrade\_suitability`, `outside\_allowed\_trading\_window`, `max\_concurrent`, `pbv2\_cap\_full`, `or\_cap\_full`, `REJECT\_SAME\_SYMBOL\_OPEN\_OVERLAP`, `max\_entries\_per\_scan`, `daily\_loss\_guard`, `risk\_cluster\_block`, `symbol\_cooloff`, `outside\_refresh\_universe`, `am\_pm\_entry\_stop`

16. EXIT設計

条件名	status	config key	現行値	入力	判定
Hard Stop -1.2%	MAINLINE_ACTIVE	discord_hard_stop_pct (Observer hard_stop_pct default 1.20)	1.20%	entry_price, mark price	price <= entry*(1-0.012) → stop_hit
Board Dynamic Trailing	MAINLINE_ACTIVE	structural_exit_policy=combined_structural_exit_v1_trailing_mfe_shadow	board_high: activate 1.0% giveback 60%; board_low: activate 0.6% giveback 40%; split@47.62	peak_pnl, pnl, entry_imbalance_percentile	peak>=activate AND pnl<=peak*giveback
No Progress Exit	MAINLINE_ACTIVE	no_progress_exit_enabled	true (linmfe_t900_i0p6_s0p05_c0p8_p0p3)	hold_sec, MFE, pnl	hold>=900s, required MFE 0.6+0.05/5m cap 0.8, pnl<0.3%
Session end exit/finalize	MAINLINE_ACTIVE	live.session_end / AM-PM runner session boundaries	AM end ~11:25; PM/session_close at session end	session clock, open positions	force close remaining → session_close
Stop Low MFE Guard	NOT_RUNTIME_REACHABLE	stop_low_mfe_guard_enabled	false	n/a	disabled
Exit Shadow Monitor (T2/T3)	NOT_RUNTIME_REACHABLE	exit_shadow_monitor_enabled / t2 / t3	false / false / false	n/a	disabled (Phase669 removed from portfolio)

- Trailing params: `trailing\_mfe\_params` L131-137
- Observer: class L162-1190
- Time bases: `observer\_entry\_time`, market entry time, `CurrentPriceTime`; stale → tag/skip per freshness_semantics_v2
- EXIT 二重防止: SafetySM / observer close idempotency（同一 position 再 close 抑制）
- PnL: `yen\_100` 正規化；MFE/MAE を event/summary に記録

- EXIT reasons: `stop\_hit`, `trailing\_mfe\_exit`, `no\_progress\_exit`, `session\_close`
- ****OFF:**** Stop Low MFE Guard / 旧 Exit Shadow Monitor = NOT_RUNTIME_REACHABLE

17. Position / Exposure管理

- max_concurrent=5, position_cap_mode=true, release=structural_exit
- OR split cap_pbv2=4 cap_or=1
- same_symbol_open_policy=no_overlap_replace

18. Intraday Refresh設計

- 10:00 AM refresh / 14:30 PM refresh (`am\_pm\_daily\_runner.AM\_REFRESH\_HHMM/PM\_REFRESH\_HHMM`)
- open position carry → fill to 50 ; registration lock + generation
- failure / exceed_cap: keep previous subscription, CONTINUE
- Code: `src/universe/intraday\_refresh.py`, `pilot\_runner` refresh path ~L6279+

19. AM/PM Session設計

- Orchestrator: `src/runner/am\_pm\_daily\_runner.py`
- AM ends ~11:25; PM screen ~12:25; trading windows YAML 準拠
- Summary preservation: `am\_pm\_summary\_preservation`

20. 同一銘柄ポリシー

`same\_symbol\_open\_policy: no\_overlap\_replace` — 保有中同一銘柄の新規 ENTRY を reject.

21. Shadow / Research設計

name	class	affects_mainline	config	runtime
I Shadow (readiness precision)	SHADOW	no	readiness_precision_shadow_enabled=true	src/small_paper/readiness_forward_shadow.py
H Shadow (readiness economics / refined H)	SHADOW	no	readiness_economics_shadow_enabled / readiness_refined_h_shadow_enabled=true	src/small_paper/readiness_forward_shadow.py
C Shadow (microsequence recovery-fail)	SHADOW	no	microsequence_recovery_fail_shadow_enabled=true	src/small_paper/microsequence_recovery_fail_forward_shadow.py
Flat Weak Range	SHADOW	no	flat_weak_range_shadow_enabled=true	src/small_paper/flat_weak_range_forward_shadow.py
NP Logger	OBSERVABILITY_ONLY	no (logger only)	np_pre_entry_feature_logger_enabled=true	src/small_paper/np_pre_entry_feature_logger.py
Sector Heat	RESEARCH_ONLY	no	n/a in production YAML	research scripts (not Monday mainline path)
Position Sizing Shadow	RESEARCH_ONLY	no	n/a production sizing fixed paper	research / live capital dry-run only
Classic Technical Indicator Research	RESEARCH_ONLY	no	classic_momentum_forward_shadow_modules	src/small_paper/classic_momentum_forward_shadow.py
Volume gate relaxation V90/V80	SHADOW	no (production remains V100)	volume_gate_relaxation_shadow_enabled=true	pilot_runner volume gate shadow
PBV2 rise5 shadow	DEPRECATED	no	pbv2_rise5_shadow_enabled=false	src/small_paper/pbv2_rise5_shadow.py
VWAP shadow reject	DEPRECATED	no	vwap_shadow_reject_enabled=false	src/small_paper/vwap_shadow_reject.py
Exit Shadow Monitor	NOT_RUNTIME_REACHABLE	no	exit_shadow_monitor_enabled=false	disabled
IHC portfolio counterfactual	SHADOW	no	runtime hook	src/small_paper/shadow_ihc_portfolio.py / ihc_shadow_counterfactual.py

22. NP Pre-entry Logger設計

- ``np_pre_entry_feature_logger_enabled=true``
- Logger only — `**no reject / no ranking**`
- ``src/small_paper/np_pre_entry_feature_logger.py``

23. Discord通知設計

W10 Router SoT: ``src/notify/discord_notification_router.py``

category	env_keys	rate_limit	fallback
TRADE_ACTUAL	KABU_SMALL_PAPER_NOTIFY_WEBHOOK_URL / KABU_SMALL_PAPER_DISCORD_WEBHOOK_URL	dedupe only	none
SESSION_SUMMARY	KABU_SMALL_PAPER_NOTIFY_WEBHOOK_URL / KABU_SMALL_PAPER_DISCORD_WEBHOOK_URL	dedupe only	none
CAP_BLOCKED	KABU_SMALL_PAPER_CAP_BLOCKED_WEBHOOK_URL	1 per symbol/session/reason via dedupe	none
OPERATIONS	KABU_DISCORD_OPERATIONS_WEBHOOK_URL	15 min	none
MARKET_CAPTURE	KABU_DISCORD_MARKET_CAPTURE_WEBHOOK_URL / KABU_MARKET_CAPTURE_WEBHOOK_URL	15 min	none
RESEARCH_SHADOW	KABU_DISCORD_RESEARCH_WEBHOOK_URL / KABU_SHADOW_DISCORD_WEBHOOK_URL	AM/PM by caller	none; no cross to TRADE
CRITICAL_SAFETY	KABU_DISCORD_CRITICAL_WEBHOOK_URL	30 min	CRITICAL_OPERATIONS_FALLBACK_DEFAULT=false

- async worker / fail-open / dedupe / retry / rate limit / HTTP 429 / audit / dead-letter
- secret masking / demo 分離 / actual/shadow 分離 / `**cross fallback 禁止**`
- webhook 未設定 → SKIP (取引は継続)
- `**Webhook URL 本体は本書に記載しない**`

24. Canonical Summary設計

``src/small_paper/canonical_summary.py`` — AM/PM summary 正規化。Shadow summary hook: ``shadow_summary_runtime_hook.py``。

25. Session Manifest / Seal設計

- SoT: ``session_seal.json`` (``w4s_seal_propagation.py``)
- pre-seal snapshot → required artifacts SHA256 + row counts → seal
- snapshot vs seal 不一致 → `SNAPSHOT_SEAL_MISMATCH` / hash mismatch
- post-seal mutation 検出で `verified=false`

26. W4S Forward Soak設計

- Module: ``research.phase687w4s_runtime_readonly_forward_soak``
- 資格: ``session_provenance=LIVE_PAPER_RUNTIME`` + `runtime_session=true`
- fixture/test/synthetic/path markers 除外 (``is_excluded_forward_path``)
- 実 Forward 3 session 蓄積待ち ; AM/PM count policy は W4S aggregate

27. SafetySM設計

- ``live_order_safety_sm_enabled=true`` but ``live_trading_enabled=false``

- DryRunBrokerAdapter / submit HARD_FAIL / write adapter absent
- actual submit=0 cancel=0 を checked runner / W4S が検証

28. Recovery設計

- Operational: `operational\_recovery.py` (disk, reconnect)
- Stateful journal: `stateful\_journal\_recovery.py`
- Assertion oracle: `recovery\_assertion\_oracle.py`

29. 例外処理設計

API/WS 例外はログ + degraded ; Discord 例外は fail-open ; Capture queue overflow は drop+metrics ; Safety 違反は fail-closed。

30. Fail-open / Fail-closed設計

領域	方針
Discord notify	fail-open
Capture writer overflow	degrade, not crash Paper
Paper precheck safety flags	fail-closed
Capture required (default)	fail-closed for Paper start
Real order path	HARD_FAIL fail-closed
open_symbols_exceed_cap	soft-open CONTINUE

31. ファイルI/O設計

JSONL append (events/push parts) 、atomic JSON status/seal、O_EXCL part create、PID file、registration lock。

32. JSONL schema一覧

詳細は `tradebot\_event\_schema.md` (paper events / capture push_part / restart_history / discord audit) 。

33. Discord schema一覧

`PAYLOAD\_SCHEMA\_VERSION=687W10.1` — `NotificationEnvelope` fields。詳細 `tradebot\_discord\_routing.md`。

34. 時刻・営業日設計

全て JST (`Asia/Tokyo`) 。trading_date=`YYYYMMDD` runtime clock。固定日付定数で本番取引日を決めない。

35. Windows起動設計

cd C:\Users\%yhach%\Documents\%tradebotfile% && .\run_paper_trade_checked.bat
PCスリープ無効・時刻同期・空き容量・Kabu起動必須。

36. テスト設計

- Runtime Gate: 323 passed (W11B 時点契約 ; `scripts/run\_runtime\_gate.py`)
- W11A regression: 91 passed
- Phase640/645: 18 passed
- compileall PASS ; strategy/canonical diff 0 ; external send 0 ; submit/cancel 0
- Gate nodes (28):

- `tests/test\_phase687w8\_paper\_trade\_checked\_runner.py`
 - `tests/test\_phase687w4t\_kabu\_readonly\_readiness.py`
 - `tests/test\_phase687w9\_market\_capture\_sidecar.py`
 - `tests/test\_phase687w11a\_monday\_p1\_fixes.py`
 - `tests/test\_am\_pm\_daily\_runner\_session\_dirs.py`
 - `tests/test\_phase549\_entry\_cluster\_guard\_runtime.py`
 - `tests/test\_phase413\_no\_overlap\_replace\_policy.py`
 - `tests/test\_canonical\_summary.py`
 - `tests/test\_phase687w7a2\_w4s\_seal\_propagation.py`
 - `tests/test\_phase687w4s\_forward\_soak.py`
 - `tests/test\_phase687w10\_discord\_notifications.py`
 - `tests/test\_phase687w10a\_shadow\_runtime\_hook.py`
 - `tests/test\_phase687\_np\_pre\_entry\_feature\_logger.py`
 - `tests/test\_phase687w7\_operational\_recovery.py`
 - `tests/test\_phase687w7a\_stateful\_recovery.py`
 - `tests/test\_phase687w7a1\_recovery\_assertion\_integrity.py`
 - `tests/test\_phase687w6\_production\_enablement\_gate.py`
 - `tests/test\_phase687w5\_kabu\_order\_contract.py`
 - `tests/test\_phase687w2\_live\_order\_safety.py`
 - `tests/test\_kabu\_register.py`
 - `tests/test\_intraday\_refresh.py`
 - `tests/test\_phase176\_intraday\_refresh\_degraded\_behavior.py`
 - `tests/test\_phase616b\_extension\_bus\_session\_end\_fix.py`
 - `tests/test\_phase662\_observer\_entry\_time\_freshness.py`
 - `tests/test\_phase684\_ihc\_counterfactual.py`
 - `tests/test\_phase650\_pbv2\_flat\_band\_shadow.py`
 - `tests/test\_allowed\_trading\_windows.py`
 - `tests/test\_position\_cap\_mode.py`
- research_long **23 files**: Monday Gate 除外 / nightly / timeout>=900s

37. Runtime Gate設計

- Manifest schema `687W11B.1` name=`monday\_active\_runtime\_gate`
- exclude_markers=['research_long']
- contract_notes={"cluster_guard": "MAINLINE_ACTIVE entry_cluster_guard_enabled=true", "stop_low_mfe": "NOT_RUNTIME_REACHABLE stop_low_mfe_guard_enabled=false", "exit_shadow_monitor": "NOT_RUNTIME_REACHABLE exit_shadow_monitor_enabled=false", "flat_band": "MAINLINE_ACTIVE pbv2_flat_band_mainline_enabled=true", "ihc": "SHADOW_ACTIVE readiness/microsequence shadows", "same_symbol": "MAINLINE_ACTIVE no_overlap_replace", "open_symbols_exceed_cap": "CONTINUE will_stop=false"}

38. セキュリティ設計

- Webhook/API password を文書・ログに出さない (redact)
- Capture に password/token/Authorization/HoldID/orders を書かない
- Real order path HARD_FAIL

39. 運用手順

詳細: `tradebot\_operations\_runbook.md`

40. 障害時対応

- Capture DEGRADED/mismatch → OPERATIONS/MARKET_CAPTURE 通知確認、当日 seal まで維持
- Paper BLOCKED / Capture CONTINUES → Paper 再起動判断、Capture 停止禁止（15:35前）
- Discord SKIP → 取引継続、Webhook env 確認
- SNAPSHOT_SEAL_MISMATCH → artifact 改変調査、W4S 非計上

41. 現在の制約

- 実注文未実装（NOT AUTHORIZED / NOT IMPLEMENTED）
- W4S 実 Forward は実市場セッション蓄積待ち
- dual WebSocket 公式保証は未確定
- Capture 実市場干渉は月曜 Forward で確認
- research_long 未完（23 files / nightly）
- repo-state 依存テストあり
- 古い legacy tests あり
- NP Logger は日数蓄積待ち
- Shadow（I/H/C, Flat Weak Range 等）は採用前
- classic technical strategy は研究中
- GO / READY は実注文許可を意味しない

42. Research Debt

- research_long 23 files 未完了/nightly
- NP Logger 日数蓄積待ち
- I/H/C / Flat Weak Range 採用前
- classic technical strategy 研究中
- repo-state 依存テスト残

43. 未実装領域

- Real order send/cancel path（NOT IMPLEMENTED）
- dual WebSocket 公式保証
- Production enablement beyond paper

44. 変更管理

- YAML 変更は `production\_config\_sha256.pin` 同期必須
- Strategy/canonical 変更は別 GO；本書は観測仕様
- OFF 機能を黙って ON にしない

45. 用語集

Term	Meaning
PBv2	momentum_volume_v2 entry path
OR overlay	Open Strength overlay entry pool
W4S	Forward soak evaluator

Term	Meaning
W10	Discord notification router
W11A	Capture registration lifetime fixes
Capture Sidecar	Independent market tape recorder
MAINLINE_ACTIVE	Affects accept/reject or lifecycle
SHADOW_ACTIVE	Logs counterfactual, no ENTRY block
NOT_RUNTIME_REACHABLE	Flag false / path unused

46. 付録

- Runtime Contract: `tradebot\_runtime\_contract.csv`
- Component inventory: `tradebot\_component\_inventory.csv`
- Config reference: `tradebot\_config\_reference.csv`
- Machine JSON: `tradebot\_current\_system\_design.json`
- Traceability: `tradebot\_traceability\_matrix.csv`
- Test matrix: `tradebot\_test\_matrix.csv`

Runtime Contract 表

Feature	Status	Current value
Entry Cluster Guard	MAINLINE_ACTIVE	True
Stop Low MFE Guard	NOT_RUNTIME_REACHABLE	False
Exit Shadow Monitor	NOT_RUNTIME_REACHABLE	False
Flat-band	MAINLINE_ACTIVE	True
I/H/C	SHADOW_ACTIVE	readiness/microsequence shadows true
same-symbol	MAINLINE_ACTIVE	no_overlap_replace
open_symbols_exceed_cap	OBSERVABILITY_ONLY	continue (will_stop=false)
Discord Router	MAINLINE_ACTIVE	W10
Registration lifetime	MAINLINE_ACTIVE	defer unregister while Capture active
live_trading_enabled	MAINLINE_ACTIVE	false
order_enabled	MAINLINE_ACTIVE	false
Hard Stop	MAINLINE_ACTIVE	1.20%
Board Dynamic Trailing	MAINLINE_ACTIVE	board_high/low tiers
No Progress Exit	MAINLINE_ACTIVE	true
OR Overlay	MAINLINE_ACTIVE	cap 4/1
NP Pre-entry Logger	OBSERVABILITY_ONLY	true
Flat Weak Range Shadow	SHADOW_ACTIVE	true
Market Capture Sidecar	MAINLINE_ACTIVE	required until 15:35
W4S Forward Soak	MAINLINE_ACTIVE	LIVE_PAPER_RUNTIME only
Real Orders	NOT_IMPLEMENTED	NOT AUTHORIZED

Safety / Real Order（結論）

live_trading_enabled=false
order_enabled=false
DryRunBrokerAdapter
submit=0
cancel=0
write adapter HARD_FAIL / absent

REAL ORDERS: NOT AUTHORIZED / NOT IMPLEMENTED
GO判定は実注文許可ではない。